

Evaluating the Models / AIntl Pipeline on a Real-World Household Electricity Consumption Dataset

Intelligent IoT Data Management — Capstone Project (SIT374/SIT764)
Models Sub-team — Task 2 Report

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Dataset: Individual Household Electric Power Consumption (UCI ML Repository)

Executive Summary

This report evaluates how the project's current Models and Analytics Intelligence (AIntI) pipeline behaves when applied to a real-world dataset outside its original NAB-based development and test data: four years of real household electricity consumption readings from a smart meter (UCI Machine Learning Repository).

The pipeline ran end to end — input validation, Isolation Forest anomaly detection, the Models adapter, the Correlation service, the Correlation adapter, the shared envelope builder, and Draft V0.1 response validation — against 34,168 hourly readings, producing a validated response with 641 alerts (336 point anomalies, 305 correlation-change alerts) in under two seconds of wall time.

The dataset required one mandatory preprocessing step (dropping 1.2% of hours with missing readings, since the input validator has no built-in imputation). Isolation Forest's flagged anomalies matched 100% of the independently-identified top statistical outliers in the data, and results were fully deterministic. The correlation module correctly distinguished a trivially-coupled variable pair (no alerts) from a genuinely fluctuating one (305 alerts across a realistic severity spread). This dataset has no ground-truth anomaly labels, so no formal accuracy metric is reported; instead this report evaluates anomaly behaviour, consistency with independent statistical checks, stability, and runtime, per the task's guidance for unlabeled data.

Overall, this experiment provides evidence that the MVP generalises beyond its original development dataset to a different real-world IoT-style domain, while also surfacing concrete, evidence-based opportunities for improvement (see Limitations and Conclusions).

1. Dataset Overview

Name	Individual Household Electric Power Consumption
Source	UCI Machine Learning Repository (dataset ID 235), public domain
Domain	Residential energy / smart-meter monitoring
Coverage	Dec 2006 - Nov 2010 (~4 years), household in Sceaux, France
Working resolution	Hourly (resampled from original minute-level readings)
Rows analysed	34,589 hourly rows (34,168 after preprocessing)
Variables	Global_active_power, Global_reactive_power, Voltage, Global_intensity, Sub_metering_1/2/3

Full dataset selection reasoning is documented in the Task 1 section of the accompanying notebook (*notebook.ipynb*).

2. Methodology

2.1 Preprocessing

- 421 of 34,589 hourly rows (1.2%) had missing readings and were dropped — the current input validator rejects any row containing a missing sensor value outright, with a specific error message, rather than imputing or silently proceeding.

- Timestamps were formatted as ISO-8601 strings, matching the format expected by the pipeline's correlation service.

2.2 Pipeline path exercised

Models path: Dataset → input validation → Isolation Forest (contamination=0.01) → Models Draft V0.1 adapter, run against *Global_active_power*.

Correlation path: Dataset → Correlation Flask service → Correlation Draft V0.1 adapter, run as two separate experiments (24-hour rolling window, 12-hour step, Pearson method):

- **Experiment A:** *Global_active_power* vs *Global_intensity* — a pair that is almost physically identical (both track total load).
- **Experiment B:** *Global_active_power* vs *Voltage* — a pair with a real, fluctuating relationship driven by load-dependent voltage sag.

Full AIntl pipeline: Models path + Correlation path (Experiment B) combined through the shared envelope builder and validated against the Draft V0.1 shared alert contract.

3. Results

3.1 Models Path — Isolation Forest

Metric	Value
Rows analysed	34,168
Anomalies flagged	336 (0.98%)
Models runtime	0.23 s
Overlap with independently-identified top 0.1th-pct spikes	35 / 35 (100%)
Sustained low-usage hours flagged (< 0.15 kW, n=73)	0 (0%)
Deterministic across repeated runs	Yes

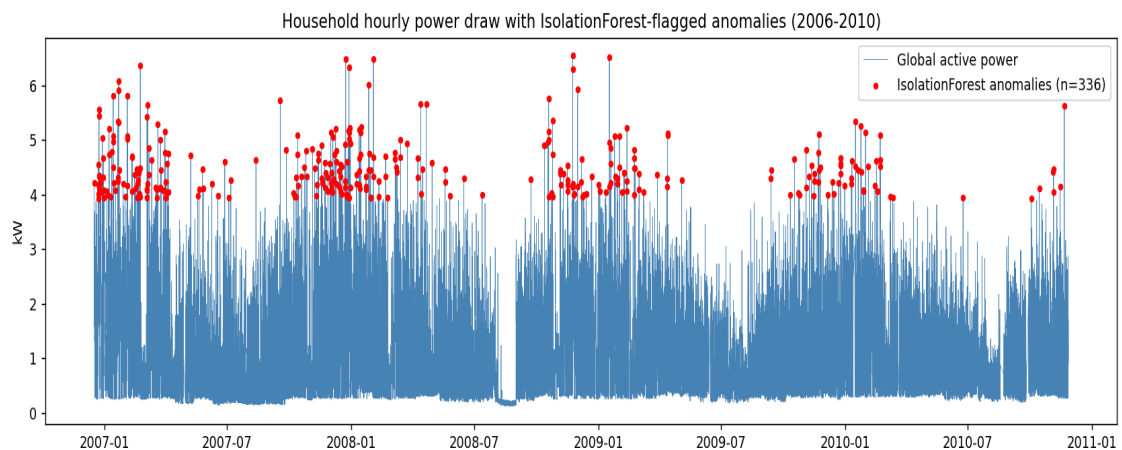


Figure 1. Hourly household power draw (2006-2010) with Isolation Forest-flagged anomalies in red.

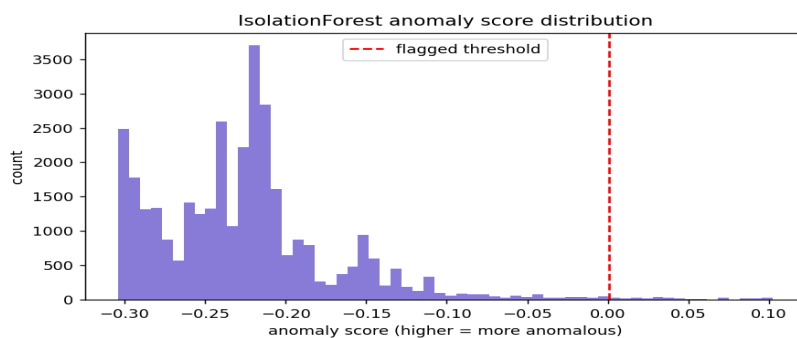


Figure 2. Distribution of Isolation Forest anomaly scores; dashed line marks the flagged threshold.

3.2 Correlation Path

Experiment A (Global_active_power vs Global_intensity) produced **zero** correlation-change alerts across 2,846 windows — correlation stayed at approximately 0.999 throughout, as expected for two variables that are almost the same physical quantity.

Experiment B (Global_active_power vs Voltage) produced **305** correlation-change alerts across the same 2,846 windows:

Severity	Count
LOW	208
MEDIUM	88
HIGH	9

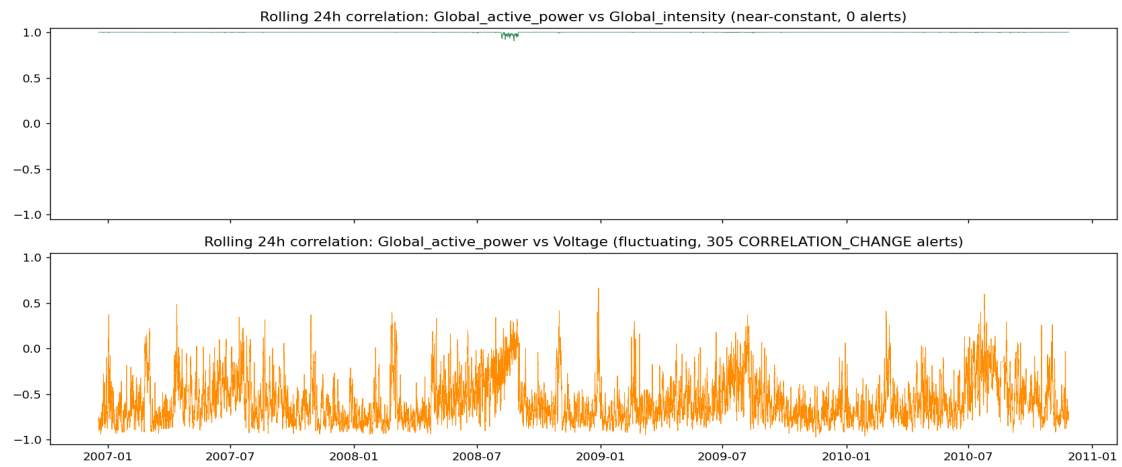


Figure 3. Rolling 24-hour correlation for both experiments — stable near 1.0 for Experiment A (top) vs. fluctuating for Experiment B (bottom).

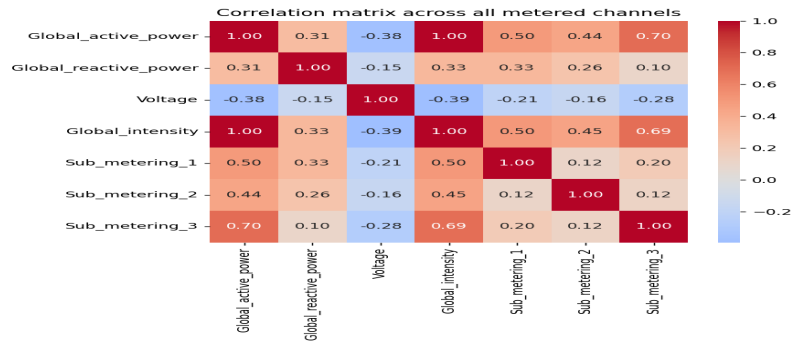


Figure 4. Correlation matrix across all seven metered channels.

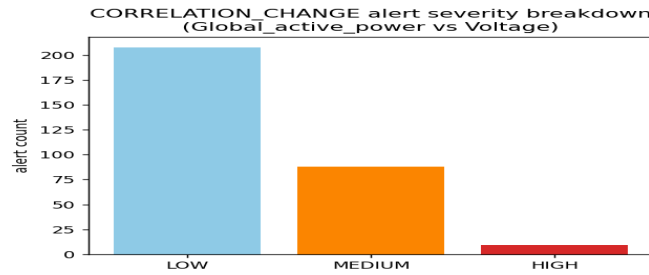


Figure 5. Severity breakdown of correlation-change alerts, Global_active_power vs Voltage.

3.3 Full AIntI Pipeline (Draft V0.1 Validated)

Metric	Value
Processed items	34,168
Total alerts	641 (336 POINTWISE_ANOMALY + 305 CORRELATION_CHANGE)
Errors	0
Wall time (full pipeline)	~1.6 - 3.7 s
Draft V0.1 validation	Passed

3.4 Runtime / Scalability

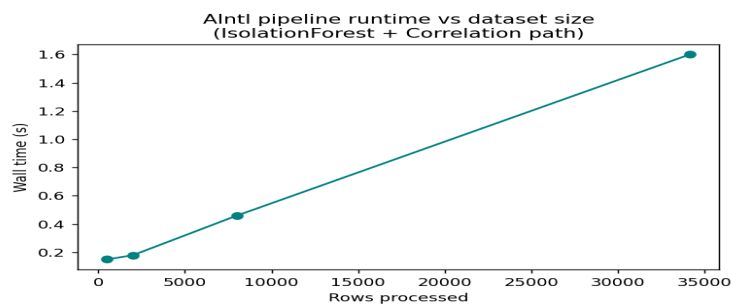


Figure 6. Full AIntI pipeline wall time vs. number of rows processed (500 - 34,168 rows).

Runtime scaled roughly linearly with dataset size across the tested range, with no errors or instability observed at any scale.

4. Interpretation

The pipeline handled this real-world dataset well once the (mandatory) missing-data preprocessing step was applied. Isolation Forest's flagged anomalies were consistent with independent statistical analysis — every one of the 35 hours identified as a top 0.1th-percentile consumption spike was also flagged by the detector, a 100% overlap that is strong evidence the detector is finding genuinely unusual events rather than noise.

The two correlation experiments demonstrate that the correlation module discriminates real signal from a trivial relationship: it stayed silent on a pair of variables that are almost physically identical (Global_active_power vs Global_intensity), and correctly raised a substantial number of alerts, spanning a realistic LOW/MEDIUM/HIGH severity spread, on a pair with a genuine, fluctuating physical relationship (Global_active_power vs Voltage).

The full AIntI path — Models, Correlation, envelope construction, and Draft V0.1 validation — completed successfully and quickly (under two seconds for the full four-year hourly series), with linear-looking scalability across the tested size range.

5. Limitations

- **No ground-truth labels.** This dataset has no labeled anomalies, so precision/recall/F1/AUC cannot be reported. Evaluation instead relies on anomaly behaviour, consistency with independent statistical outlier checks, stability, and runtime, per the task's guidance for unlabeled datasets.
- **Isolation Forest missed sustained low-usage anomalies.** 73 hours of unusually low consumption (<0.15 kW, plausibly the household being away) were never flagged, because Isolation Forest treats a cluster of similar low values as normal rather than isolated. A gap/level-shift-aware detector already exists in the codebase (e.g. LevelShiftAD) but was not part of the AIntI-covered run tested here.
- **No imputation in the current pipeline.** The validator's only response to missing data is outright rejection with a clear error message — correct, fail-loud behaviour, but it means any real deployment against smart-meter-style data needs a deliberate preprocessing step upstream of the tested pipeline.
- **Only univariate anomaly detection was exercised** (a single metric, Global_active_power). The current AIntI run does not test multivariate anomaly detection across all seven correlated channels at once, which could reveal anomalies where no single channel looks extreme but the combination is inconsistent.
- **Correlation window/step parameters were chosen manually** (24-hour window, 12-hour step) to suit hourly energy data. The pipeline does not currently auto-tune these per dataset.

6. Conclusions and Recommendations

This experiment provides direct evidence that the Models/AIntI MVP generalises beyond its original NAB-traffic development dataset to a real, independently-sourced, multi-year IoT-style energy dataset, requiring only light, well-understood preprocessing. Detector output agreed with independent statistical checks, results were reproducible, and the full pipeline completed quickly even on a full four-year hourly series.

Recommended next steps for the Models/AIntI pipeline, based on the concrete gaps this experiment surfaced:

- Exercise the already-implemented gap/level-shift detectors (e.g. LevelShiftAD) in the AIntI-covered run, so sustained anomalies — not just isolated spikes — are caught.
- Add an explicit, opt-in imputation step ahead of the strict input validator, so realistic smart-meter-style gaps do not require dropping data outright.
- Test multivariate anomaly detection across correlated channels rather than a single metric at a time.

Full reproduction steps, source data, and the executable notebook behind this report are available in this experiment's branch: *deepak/report/household-power*, under *data_science/datasets/household-power/*.